

RESEARCH INTEREST

My research focuses on Semantic Web, Knowledge Graphs, Neuro-Symbolic AI, Large Language Models (LLMs), Continual Learning, Retrieval-Augmented Generation (RAG), and Trustworthy AI. I develop scalable, explainable, and continually learning methods for knowledge graph construction, completion, reasoning, and information extraction in the Generative AI era. My research combines symbolic reasoning with foundation models and is applied to scientific knowledge management, including biodiversity knowledge integration.

EXPERIENCE

02/2026 – Present **Research Scientist** Senckenberg – Leibniz Institution for Biodiversity and Earth System Research | Germany (Full-time)

- Developed knowledge graph and LLM-based approaches for semantic integration, entity resolution, and biodiversity knowledge modeling within large-scale scientific data infrastructures.
- **Secured a prestigious EuroHPC GPU grant**, unlocking high-performance computing infrastructure to scale deep learning models and accelerate large-scale text-extraction workloads.

02/2023 – 03/2025 **Research Associate** Technische Universität Berlin | Berlin, Germany (Full-time)

- **Engineered core digital infrastructure** for the Berlin University Alliance (BUA) project, alongside *NFDI4DataScience* and *NFDI4Cat* initiatives.
- **"A Digital Research Space"**, a centralized web portal (**meta4bua**) that successfully federates diverse institutional repositories across BUA.
- **Designed and developed the unified metadata portal** for NFDI4DataScience and NFDI4Cat to optimize cross-disciplinary discovery.
- **Tech Stack:** Java, Python, Kotlin, SPARQL, RDF, OWL, Ontologies, Virtuoso, Elastic, Vue.js, Docker, Large Language Models (LLMs), and federated database systems.
- **Teaching:** Open Distributed Systems Project Class, TU Berlin (Winter 2023). Advanced Research Data Infrastructure Project Class, TU Berlin (Summer 2024)

10/2016 – 03/2017 **Software Engineer** MetGlobal | Istanbul, Turkey (Full-time)

- **Development and maintenance** of call center CRM architectures alongside optimized analytics dashboards using Google Charts for cross-functional teams.
- **Tech Stack:** PHP, JetBrains, MySQL, JavaScript, Vagrant.

02/2016 – 08/2016 **Software Engineer** Garanti BBVA Technology | Istanbul, Turkey (Full-time)

- **Development and maintenance** of core Garanti BBVA Emeklilik applications, CRM portals, and a complete system re-architecture of the consumer corporate pensions platform.
- **Tech Stack:** MS Dynamics CRM Development, Jenkins, T-SQL, MS SQL Server, Eclipse, Visual Studio, Java Spring Boot, C#, Java EE, TFS, SVN.

07/2014 – 08/2014 **Student Intern** Multimedia Databases Laboratory, METU | Ankara, Turkey

- **Graph Database Modeling:** Modeled complex video data networks on graph configurations using Neo4j and benchmarked query evaluations against structural object databases (db4o).

EDUCATION

12/2021 – Present **PhD in Computer Science** Free University of Berlin | Germany

Thesis: *Continual Relation Extraction for Knowledge Graph Completeness*

Expertise: Transformers, LLMs, Continual Learning (Task-Incremental, Meta-Learning), Encoder-Decoder Models, PyTorch, Relation Extraction, Scikit-learn, HuggingFace, SpaCy.

Supervisor: Prof. Dr. Adrian Paschke

Note: Thesis submitted and defense in September.

10/2018 – 09/2021 **Master of Science in Data Science** University of Potsdam | Germany

Thesis: *A Deep Learning Approach for Domain-Specific Automated Ontology Construction*

- **Semantic Technologies:** SpaCy, Flair, YAGO, DBpedia, Wikidata, Owlready2, PyTorch.
- **Methodologies:** NLP, Named Entity Recognition, Graph Attention Networks, and Representation Learning.

09/2010 – 07/2015 **Bachelor of Computer Engineering** Ege University | Izmir, Turkey

Thesis: *Testing Ontology-Based Semantic Relatedness Measurement in Graph Database in Flickr Dataset.*

09/2005 – **Demre Anatolian High School**

06/2009

Math and Science Track

Antalya, Turkey

AWARD AND GRANT

6/2023	Best Scientific Research Report - Awarded at the International Semantic Web Research Summer School (ISWS) 2023. - Core iteration accepted and presented at the Wikidata Workshop at ISWC 2023.	Award
3/2023	ISWS Summer School Grant Awarded a full competitive grant covering all travel and lodging expenses (Bertinoro, Italy).	Grant
9/2020	Mediterranean Machine Learning School Grant Selected candidate for fully subsidized registration to the MMLS intensive ML curriculum.	Grant
8/2019	15th Web Reasoning Summer School Grant Received full operational grant funding for school tenure at the Free University of Bolzano, Italy.	Grant
10/2017	Graduate Education Scholarship for Study Abroad - Highly competitive full fellowship awarded by the Republic of Turkey Ministry of National Education. - Financed all institutional fees and living expenses for both the MSc and PhD tracks.	Scholarship
7/2015	Honor Student Graduated with official Honor standing from Ege University based on cumulative GPA.	Award
09/2005 06/2009	-High School Entrance Scholarship Merit scholarship awarded by the Ministry of National Education matching top performance tiers.	Scholarship

SELECTED CERTIFICATES

10/2023	Aya Open Science Project Cohere4AI Achieved 1000 Contribution Points to "Aya: An Open Science Initiative to Accelerate Multilingual AI Progress".	Certificate
10/2023	Generative AI with Large Language Models Credential verified by DeepLearning.AI; covered advanced instruction on LLMs, SFT, and NLP scaling paths.	Certificate
6/2023	International Semantic Web Research Summer School (ISWS) Theme: Creative AI. Courses: KR Querying/Reasoning For Large KGs, FAIR Knowledge Graphs, Building Large KGs.	Certificate
9/2020	Mediterranean Machine Learning School Advanced coursework completed in Natural Language Processing and Geometric Deep Learning Models.	Certificate
8/2019	15th Reasoning Web Summer School 2019 Theme: Explainable AI. Core Focus: Knowledge Representation, Rule Mining in KBs, Formal Concept Analysis, and ASP.	Certificate
7/2019	Advanced Course in AI Summer School Hosted by the Technical University of Crete, Greece. Focused on Multi-Agent Planning under Uncertainty and Learning.	Certificate

SELECTED PUBLICATIONS

	Reversing Arrows: A Benchmark Dataset for Inverse Relation Directionality in LLMs*	Conference (under review)
JWS	SHACL-Based Neuro-Symbolic Explainability for Continual Relation Extraction and Hallucination Resolution*	Journal Article (under revision)
IEEE Access	Large Language Models for Continual Relation Extraction* Applied Research Track	Journal Article
SWJ	Retrieval-Augmented Generation-based Relation Extraction* Special Issue about Knowledge Graph Generation for Semantic Web Journal (SWJ)	Journal Article
ACL 2025	Fine-Tuning Large Language Models for Relation Extraction within a Retrieval-Augmented Generation Framework*	XLLM Workshop Paper

ESWC 2024	Converter: Enhancing Interoperability in Research Data Management* Extended Semantic Web Conference 2024	Main Conference-Poster paper
Semantics '24	Exploring Prompt Generation Utilizing Graph Search Algorithms for Ontology Matching Semantics Conference 2024	Conference Paper
ISWC 2023	Draw Me Like My Triples: Leveraging Generative AI for Wikidata Image Completion Wikidata'23 Workshop, collocated with ISWC 2023, Athens, Greece.	Inproceedings
SWAT4HCLS	Fine-Grained Named Entities for Corona News* The 14th International SWAT4HCLS Conference, Basel, Switzerland.	Inproceedings
ISWC 2022	GraphMatcher: A Graph Representation Learning Approach for Ontology Matching* International Workshop on Ontology Matching, ISWC-2022.	Inproceedings
Springer	Hybrider KI (Hybrid AI) Part 1: Machine Learning for knowledge-based AI — Chapter "Unterstützung klinischer Studien mit hybrider KI"	Book Chapter

SERVICES

2026	ISWC 2026 Resource Track	Reviewer
2026 – Pres.	Editorial Activities	Journal of Web Semantics

VOLUNTEERING

03/2023 09/2023	-Volunteer Mentor Disaster Mentorship Program Organized by SistersLab - Association for Women in Science and Technology
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LANGUAGE

English — C1, **Turkish** — Native

* First author